

Spectral Trace Formula and Smoothed Zero Sums:

A Prime–Zero Duality Framework

Ulrich Tehrani

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Abstract

We study a one-parameter family of finite-rank operators $\tilde{T}(\sigma)$ acting on a finite-dimensional Hilbert space H_{null} over the imaginary parts of the non-trivial zeros of $\zeta(s)$. The operators arise from an explicit coupling map $\Phi(\sigma): H_{\text{str}} \rightarrow H_{\text{null}}$ encoding prime–zero resonance via $\sin(\sigma\gamma_k \log p)$. No hypothetical input is used: no Riemann Hypothesis, no Hilbert–Pólya postulate, no GUE conjecture.

What is proved. We establish an exact trace formula $\text{Tr}(\tilde{T}(\sigma)) = D_{\text{SEL}} - O(\sigma)$ algebraically, for all $\sigma \in \mathbb{R}$, by a two-line trigonometric computation (Theorem 3.1). The decomposition $B = \sum_{p \leq \kappa} (\log p)^2 \text{Re}(Z_p^{(2)})$ is proved from first principles (Proposition 5.4). These results require no analytic assumptions.

What is numerical. The limiting energy asymmetry satisfies numerical extrapolation gives $\eta_{\text{ren}}(\kappa) \rightarrow 0.81$ on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ [limit not proved, Observation 4.1]. Three spectral signatures at $\sigma = \frac{1}{2}$ — a trace spike of +47%, a dominant-eigenvalue maximum, and a spectral-gap minimum — are observed numerically [Numerical, §2]. The γ^2 -weighted curvature-bias quantity satisfies $B = -19342.5 < 0$ at $\kappa = 53$, $\varepsilon = 0.05$, $N = 100$ [Numerical].

What remains open. The Bias Conjecture — for each fixed prime p , $Z_p^\infty(\varepsilon) < 0$ for all sufficiently small $\varepsilon > 0$ — is the central open problem (Conjecture 5.7). Its proof, together with Proposition 5.8, would establish the sign of the unweighted integrated bias B_{int} . Closing the curvature-bias argument for $O''(\frac{1}{2}) = -2B$ additionally requires the Weighted Bias Bridge from B_{int} to the γ_k^2 -weighted quantity B , or an independent proof that $B < 0$; the sign transfer to B_{int} is additionally subject to the auxiliary truncation-error condition of Proposition 5.8. The finite-cutoff identity $\eta_{\text{ren}}(\kappa) = 1 - m_1(\nu_\kappa^{(w)})$ holds exactly; convergence of the weighted limiting spectral measure $\nu_\infty^{(w)}$ and identification of its limiting first moment are open [Open, §4].

Structure. The proved and conditional statements in §§2–5 are non-circular and status-separated. No use of the Riemann Hypothesis, the GUE conjecture, the Montgomery pair correlation conjecture, or the Hilbert–Pólya postulate is made anywhere in §§2–5.

Notation and Conventions

Critical line. Throughout the series, the critical line means $\Re(s) = \frac{1}{2}$. The parameter σ denotes the real part of s , so the critical line corresponds to $\sigma = \frac{1}{2}$. The reference point $\sigma_0 = \frac{1}{2}$ is chosen as the object of study, not assumed to be the unique zero location.

Global parameters. Unless otherwise stated, computations use the reference parameters $\kappa = 53$ (prime cutoff, giving $\pi(\kappa) = 16$ active primes), $\varepsilon = 0.05$ (Gaussian damping parameter), $N = 100$ (number of Riemann zero ordinates), and $\sigma_0 = \frac{1}{2}$. The zero ordinates $\gamma_1 < \gamma_2 < \dots < \gamma_N$ are taken as fixed numerical inputs; the last ordinate is $\gamma_{100} = 236.524$.

Main objects. The coupling map $\Phi(\sigma): H_{\text{str}} \rightarrow H_{\text{null}}$ is defined by

$$\Phi(\sigma)_{k,p} := e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\sigma \gamma_k \log p).$$

The operator family $\tilde{T}(\sigma) := \Phi(\sigma) \circ \Phi(\sigma)^*$ acts on H_{null} and satisfies the trace formula $\text{Tr}(\tilde{T}(\sigma)) = D_{\text{SEL}} - O(\sigma)$, where $D_{\text{SEL}} := \frac{1}{2} A(\varepsilon, N) \cdot \pi(\kappa) = 10.985$ is σ -independent and $O(\sigma)$ carries the full σ -dependence:

$$O(\sigma) := \frac{1}{2} \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} \cos(2\sigma \gamma_k \log p).$$

The curvature-bias quantity is $B := \sum_{p \leq \kappa} (\log p)^2 \text{Re}(Z_p^{(2)})$, where the γ_k^2 -weighted phasor sum is

$$Z_p^{(2)} := \sum_{k=1}^N \gamma_k^2 e^{-\varepsilon^2 \gamma_k^2} p^{i\gamma_k},$$

the unweighted phasor sum is $Z_p := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} p^{i\gamma_k}$ (Definition 5.1), and the Guinand–Weil object is $\tilde{Z}_p^{\text{GW}}(\varepsilon) := \sum_{\rho} h_{p,\varepsilon}^{\text{even}}((\rho - \frac{1}{2})/i)$.

Weights. The energy asymmetry η_{ren} recalled in § 4 uses the weight vector $c_p = c_p^{\text{ren}} := \sqrt{f_p}$, where $f_p := 4(\log p)^2(2p-1)/(p(p-1)^2)$ (Paper 4’s convention), giving $\eta_{\text{ren}}(53) = 0.66927$. This specialisation differs from Paper 3’s η_{orig} , which uses $c_p^\eta := \sqrt{V_p(\frac{1}{2})}$ and gives the numerically different value 0.69078; see Paper 3, Notation and Conventions.

Not to be confused with. The trace-formula constant $D_{\text{SEL}} = 10.985$ (this paper) is distinct from the diagonal energy $D \approx 9.470$ of Paper 2; the subscript SEL is mandatory. The weighted sum $Z_p^{(2)}$ (Definition 5.3, carries γ_k^2 , enters the curvature decomposition of B) is distinct from the unweighted sum Z_p (Definition 5.1, enters the Bias Conjecture). The finite truncation $Z_p(\varepsilon)$ (Definition 5.1) is the half-sum over positive ordinates $k = 1, \dots, N$. Its real part $Z_{p,N}^+(\varepsilon) := \text{Re } Z_p(\varepsilon)$ is the object used in Proposition 5.8. The infinite analogue $Z_p^\infty(\varepsilon) := 2 \sum_{k=1}^\infty e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p)$ is the object of the Bias Conjecture; $Z_{p,N}^+ = \frac{1}{2}(Z_p^\infty + R_{p,N}^\infty)$ with tail $|R_{p,N}^\infty| \lesssim 10^{-61}$ at $N = 100$, $\varepsilon = 0.05$. The full Guinand–Weil sum $\tilde{Z}_p^{\text{GW}}$ (all zeros of ζ) provides formal motivation via the explicit formula but is not used in computations; see Definition 5.2. The curvature sum $B = -19\,342.5$ (which includes γ_k^2 weights) is distinct from the integrated bias $B_{\text{int}} = -42.21$ (which does not). The constant $C_T \approx 2\pi(\kappa)$ governing eigenvalue decay must not be confused with $C_\eta \approx 0.39$ governing the renormalized spectral radius.

1 Introduction

1.1 Background and Motivation

This paper is the fifth in a series developing an operator-theoretic framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The series is built around a finite-dimensional Hilbert-space model in which prime numbers mediate coupling between zeros, encoded in an explicit operator family.

Paper 1 [1] introduced the local curvature function $H_{\text{local}}(\sigma, \kappa) := \psi_1(\sigma) + \sum_{p \leq \kappa} V_p(\sigma)$ (see **Paper 1**, Notation and Conventions), where each summand $V_p(\sigma) > 0$ is a per-prime contribution. A key finding was the divergence $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 2(\log \kappa)^2 \rightarrow \infty$, contrasted with convergence $H_{\text{local}}(\sigma, \kappa) \rightarrow C(\sigma) < \infty$ for all $\sigma > \frac{1}{2}$, identifying the critical line as a phase boundary observed, not postulated.

Paper 2 [2] connected this divergence to the Weil explicit functional, constructing admissible test functions for which the curvature divergence becomes a structural feature of the Weil energy.

Paper 3 [3] made the geometric structure explicit: two finite-dimensional Hilbert spaces H_{str} (over primes) and H_{null} (over zeros) connected by a linear map Φ , with loop operator $T = \Phi^* \circ \Phi$. The energy asymmetry $\eta_{\text{orig}} > 0$ was established numerically for the Paper 3 weight $c_p^\eta = \sqrt{V_p(1/2)}$. The convergent renormalised weight scale $f_p = 4(\log p)^2(2p - 1)/(p(p - 1)^2)$, used later for η_{ren} , belongs to the Paper 2/Paper 4 convention.

Paper 4 [4] introduced the dual operator $\tilde{T} = \Phi \circ \Phi^*$, acting on H_{null} , and established a conditional analytic proof that $\eta_{\text{ren}} > 0$ under the remainder control assumption (E_{rem}); see Paper 4, § 5. Non-vanishing on $\text{Re}(s) = 1$ is settled unconditionally by Hadamard [9] and de la Vallée Poussin [10] (1896); the genuinely open condition is the remainder estimate for the Weyl-equidistribution contribution.

The present paper takes a different route. Rather than pursuing the equidistribution condition of Paper 4, we establish an exact trace formula for $\tilde{T}(\sigma)$ — purely algebraic, requiring no analytic assumptions — and introduce the smoothed zero-sum theory that exposes a structural negative bias at $\sigma = \frac{1}{2}$. The central open problem is the **Bias Conjecture**: for each fixed prime p , $Z_p^\infty(\varepsilon) < 0$ for all sufficiently small $\varepsilon > 0$. Its proof would complete the $\sigma = \frac{1}{2}$ Curvature Bias argument via the Reduction Lemma, subject to the stated auxiliary conditions. **We do not prove the Bias Conjecture in this paper.**

Relation to other approaches. Connes and Consani [14] construct an archimedean compression of the scaling action to Sonin’s space, establishing a trace formula for the Weil functional at the archimedean place via an infinite-dimensional projection-based operator. The present paper’s finite-rank operator $\tilde{T}(\sigma) = \Phi(\sigma)\Phi(\sigma)^*$ on H_{null} is indexed by explicit Gaussian-damped sine phasors; it is a Gram-type construction, not a compression of an adèlic action.

Connes, Consani, and Moscovici [17] construct a self-adjoint operator as a rank-one perturbation of the scaling spectral triple, achieving numerical alignment with the first fifty zero ordinates. The present construction differs structurally: $\tilde{T} = \Phi\Phi^*$ is a Gram-type operator without spectral

postulates and without rank-one perturbation structure. The zero ordinates γ_k are external numerical inputs, not spectrum.

Yakaboylu [18] constructs a non-symmetric operator R on a Hilbert space whose spectrum is determined by the zeros of $\zeta(s)$ via Bombieri’s quadratic functional, invoking the Hilbert–Pólya pathway. The present paper does not postulate any operator whose spectrum equals the zero ordinates; $\tilde{T}$ is explicitly falsified as a Hilbert–Pólya operator in § 2.2.

1.2 Main Results

The paper contains three main components.

The first is Theorem 3.1, an exact trace formula $\mathrm{Tr}(\tilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$, proved algebraically for all $\sigma \in \mathbb{R}$. Here D_{SEL} is an explicit σ -independent constant and $O(\sigma)$ is a finite oscillatory sum over primes and zeros; the entire σ -dependence of the trace resides in $O(\sigma)$.

The second is a numerical observation (Observation 4.1) that $\eta_{\mathrm{ren}}(\kappa) \rightarrow 0.81$ on the tested grid, motivating a weighted limiting spectral measure $\nu_\infty^{(w)}$ with $\eta_\infty = 1 - m_1(\nu_\infty^{(w)})$ (the finite-cutoff identity uses $\nu_\kappa^{(w)}$, not the unweighted empirical measure ν_κ ; convergence open). The numerical content is robust for $\kappa \leq 1009$; the convergence and tightness of $\nu_\kappa^{(w)}$ and the identification of its limiting first moment remain open.

The third is the smoothed zero-sum framework of § 5. Proposition 5.4 decomposes the curvature quantity as $B = \sum_p (\log p)^2 \mathrm{Re}(Z_p^{(2)})$, and the Structural Reduction (§ 5.2) formally isolates a candidate main term that is strictly negative via the Guinand–Weil explicit formula. The Bias Conjecture (Conjecture 5.7) is stated as the central open problem, and Proposition 5.8 records the conditional sign transfer.

1.3 Reader Contract

This paper *proves* unconditionally the exact trace formula $\mathrm{Tr}(\tilde{T}(\sigma)) = D_{\mathrm{SEL}} - O(\sigma)$ (Theorem 3.1) and the curvature decomposition $B = \sum_p (\log p)^2 \mathrm{Re}(Z_p^{(2)})$ (Proposition 5.4). Proposition 5.8 is conditional on the Bias Conjecture and the stated truncation-error condition, and it transfers the sign only to the unweighted integrated bias B_{int} , not to the γ_k^2 -weighted curvature quantity B .

This paper *establishes numerically* that $\eta_{\mathrm{ren}}(\kappa)$ appears to approach 0.81 on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ [numerical extrapolation; no convergence theorem] (Observation 4.1), that $\mathrm{Re}(Z_p(\varepsilon)) < 0$ for 14 of the 16 primes $p \leq 53$ at $\varepsilon = 0.05$ (exceptions: $p = 37$ and $p = 53$), and that $B_{\mathrm{int}} = -42.21 < 0$ at reference parameters.

This paper *leaves open* the Bias Conjecture (Conjecture 5.7), the tightness and convergence of $\nu_\kappa^{(w)}$ and, if a limiting measure $\nu_\infty^{(w)}$ exists, the limiting identity $\eta_\infty = 1 - m_1(\nu_\infty^{(w)})$, as well as analytical control of the Γ -term in the Guinand–Weil decomposition.

2 The Operator Family $\tilde{T}(\sigma)$

2.1 Hilbert Spaces and Coupling Map

Let $\kappa > 1$ be a prime cutoff and $N \geq 1$ a zero cutoff. Write $\{p_1 < p_2 < \cdots < p_P\}$ for the primes up to κ , and $\gamma_1 < \gamma_2 < \cdots < \gamma_N$ for the imaginary parts of the first N non-trivial zeros of $\zeta(s)$, listed in increasing order and taken as fixed numerical input parameters. The reference parameters used throughout are $\kappa = 53$, $\varepsilon = 0.05$, $N = 100$, $\sigma_0 = \frac{1}{2}$.

Definition 2.1 (Hilbert spaces). Set $H_{\text{str}} := \ell^2(\{p \text{ prime} : p \leq \kappa\})$ (of dimension P) and $H_{\text{null}} := \ell^2(\{\gamma_k : k = 1, \dots, N\})$ (of dimension N), each equipped with the standard inner product.

Remark. The values γ_k are used as numerical inputs throughout. No property of their distribution — in particular, no assertion that $\text{Re}(\rho_k) = \frac{1}{2}$ for the corresponding zeros ρ_k — is assumed or used in any proof.

Definition 2.2 (Coupling map and operator family). Let $\varepsilon > 0$. Define $\Phi(\sigma): H_{\text{str}} \rightarrow H_{\text{null}}$ by

$$\Phi(\sigma)_{k,p} := e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\sigma \gamma_k \log p),$$

and set

$$\tilde{T}(\sigma) := \Phi(\sigma) \circ \Phi(\sigma)^*: H_{\text{null}} \rightarrow H_{\text{null}}.$$

Explicitly, $\tilde{T}(\sigma)_{kl} = \sum_{p \leq \kappa} e^{-\varepsilon^2 (\gamma_k^2 + \gamma_l^2) / 2} \sin(\sigma \gamma_k \log p) \sin(\sigma \gamma_l \log p)$. The reference operator is $\tilde{T} := \tilde{T}(\frac{1}{2})$.

The operator $\tilde{T}(\sigma)$ is self-adjoint and positive semi-definite for all σ , since it is of the form $\Phi\Phi^*$. Its rank is at most P . We write $\mu_1(\sigma) \geq \mu_2(\sigma) \geq \cdots \geq \mu_N(\sigma) \geq 0$ for its eigenvalues in non-increasing order.

2.2 Spectral Properties

$\tilde{T}$ is not a Hilbert–Pólya operator. The Hilbert–Pólya conjecture postulates a self-adjoint operator whose eigenvalues are precisely the imaginary parts γ_k of the non-trivial zeros. $\tilde{T}$ is not such an operator. Numerical computation ($\kappa = 53$, $\varepsilon = 0.05$, $N = 100$) gives $\mu_j \sim C_T / \gamma_{k(j)}$ with $C_T \approx 2\pi(\kappa)$ and Pearson correlation $r_1 = 0.950$, where $k(j)$ denotes the dominant zero-index of eigenmode j . On the tested finite list, the observed eigenvalues follow an inverse-ordinate decay pattern $\mu_j \approx C_T / \gamma_{k(j)} \rightarrow 0$, which is structurally incompatible with the Hilbert–Pólya requirement $\mu_j = \gamma_j \rightarrow \infty$. $\tilde{T}$ is a resonance operator: $\tilde{T}_{kl}$ measures the prime-mediated coupling between zeros γ_k and γ_l . The term *resonance* is used here in the descriptive sense of oscillatory prime-mediated coupling, not in the spectral sense of Connes [11], where noncritical zeros appear as resonances of an adèlic action.

The constant $C_T \approx 2\pi(\kappa)$ governing the eigenvalue decay is distinct from $C_\eta \approx 0.39$, which governs the growth of the maximum eigenvalue of the renormalized operator $\lambda_{\text{max}}(T_{\text{ren}}) \approx C_\eta \cdot \pi(\kappa)$ (numerically observed in Paper 4 [4]). These two constants must not be confused.

Three numerical signatures at $\sigma = \frac{1}{2}$. The following spectral features are observed numerically at $\sigma = \frac{1}{2}$ ($\kappa = 53$, $\varepsilon = 0.05$, $N = 100$). First, the trace satisfies $\text{Tr}(\tilde{T}(\frac{1}{2})) = 14.624$, approximately 47% above the minimum observed on the sampled local window $\sigma \in [0.3, 0.7]$. Second, the dominant eigenvalue $\mu_1(\sigma)$ attains its maximum $\mu_1(\frac{1}{2}) = 6.835$ at $\sigma = \frac{1}{2}$. Third, the spectral gap $|\mu_8(\frac{1}{2}) - \mu_9(\frac{1}{2})| = 0.0019$ attains its minimum at $\sigma = \frac{1}{2}$. These are numerical observations, not theorems.

3 The Trace Formula

Theorem 3.1 (Trace Formula, proved algebraically). *For all $\sigma \in \mathbb{R}$,*

$$\text{Tr}(\tilde{T}(\sigma)) = \sum_{j=1}^N \mu_j(\sigma) = D_{\text{SEL}} - O(\sigma),$$

where

$$D_{\text{SEL}} := \frac{1}{2} A(\varepsilon, N) \cdot \pi(\kappa), \quad A(\varepsilon, N) := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2},$$

$$O(\sigma) := \frac{1}{2} \sum_{k=1}^N \sum_{p \leq \kappa} e^{-\varepsilon^2 \gamma_k^2} \cos(2\sigma \gamma_k \log p).$$

The quantity D_{SEL} is σ -independent. The entire σ -dependence of the trace resides in $O(\sigma)$.

Proof. From Definition 2.2,

$$\text{Tr}(\tilde{T}(\sigma)) = \sum_{k=1}^N \tilde{T}(\sigma)_{kk} = \sum_{k=1}^N \sum_{p \leq \kappa} e^{-\varepsilon^2 \gamma_k^2} \sin^2(\sigma \gamma_k \log p).$$

Applying $\sin^2(x) = (1 - \cos 2x)/2$ to each term:

$$= \frac{1}{2} \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} - \frac{1}{2} \sum_{k,p} e^{-\varepsilon^2 \gamma_k^2} \cos(2\sigma \gamma_k \log p) = D_{\text{SEL}} - O(\sigma). \quad \square$$

The proof is purely algebraic: no analytic assumptions, no asymptotic arguments, no input from the Riemann Hypothesis. The formula is exact for any finite κ , N , ε , σ .

Numerical verification ($\kappa = 53$, $\varepsilon = 0.05$, $N = 100$, σ -grid) gives $\max_{\sigma} |\text{Tr}_{\text{spec}}(\sigma) - (D_{\text{SEL}} - O(\sigma))| < 10^{-14}$, confirming the identity at floating-point precision. The reference values are: $A(\varepsilon, N) = 1.37306$; $\pi(53) = 16$; $D_{\text{SEL}} = 10.9845$; $O(\frac{1}{2}) = -3.6391$; $\text{Tr}(\tilde{T}(\frac{1}{2})) = 14.624$.

Corollary 3.2 (Extrema correspondence, proved). *The functions $\sigma \mapsto \text{Tr}(\tilde{T}(\sigma))$ and $\sigma \mapsto -O(\sigma)$ differ by the constant D_{SEL} ; equivalently,*

$$\text{Tr}(\tilde{T}(\sigma)) + O(\sigma) \equiv D_{\text{SEL}}.$$

In particular, maxima of $\text{Tr}(\tilde{T}(\sigma))$ correspond to minima of $O(\sigma)$, and conversely.

Numerically, $\sigma = \frac{1}{2}$ is a local maximum of $\text{Tr}(\tilde{T}(\sigma))$, equivalently a local minimum of $O(\sigma)$. This is a numerical observation.

Remark (Positioning relative to classical trace formulas). *The decomposition $\text{Tr}(\tilde{T}(\sigma)) = D_{\text{SEL}} - O(\sigma)$ has a structural parallel to the Selberg trace formula, in which a spectral sum equals a sum of geometric data: D_{SEL} plays the role of the constant geometric term and $O(\sigma)$ plays the role of the oscillatory prime-geodesic sum. This analogy is motivational; no formal connection to the Selberg trace formula is claimed.*

The distinction should be stated precisely. The classical Selberg trace formula matches Laplace spectra on hyperbolic surfaces with lengths of closed geodesics; the Arthur trace formula balances geometric and spectral distributions for reductive groups; the Connes trace formula [11] interprets the explicit formula on the adèle-class space with zeros as a spectral absorption set. In Burnol's Fourier-analytic approach [12], the explicit formula is recast in operator-theoretic language via a conductor operator $H = \log|x| + \log|y|$. By contrast, the identity in Theorem 3.1 is a finite-rank algebraic trace computation for an explicitly constructed prime-zero Gram family; it does not arise from group actions, orbital integrals, or an adèlic framework.

Three further points deserve emphasis. Theorem 3.1 makes no assertion about $\sigma = \frac{1}{2}$ specifically; the trace formula is uniform in σ . No connection to the Riemann Hypothesis follows from Theorem 3.1 alone. Finally, $\text{Tr}(\tilde{T}(\sigma)) \neq \Gamma_\kappa(\sigma)$ in general: the Pearson correlation is +0.9915 (numerical), but the ratio is not constant (coefficient of variation 54.76%), owing to a $(\log p)^2$ discrepancy in the definitions, since $\tilde{T}$ uses $\sin(\sigma\gamma_k \log p)$ while Γ_κ involves $(\log p)^2$ factors explicitly.

The symbol D_{SEL} carries a mandatory subscript to distinguish it from the Paper 2 diagonal energy scalar $D := \sum_p f_p \approx 9.470$; these two quantities are unrelated.

4 Asymptotic Spectral Measure

We recall the energy asymmetry from Papers 3 and 4. For prime cutoff κ ,

$$\eta_{\text{ren}}(\kappa) := 1 - \frac{c^T G^{\text{un}} c}{E_{\text{str}}(\kappa)},$$

where $c = (c_p)_{p \leq \kappa}$ is the canonical weight vector with $c_p := \sqrt{4(\log p)^2(2p-1)/(p(p-1)^2)}$; G^{un} is the full unnormalised Gram matrix of the resonance vectors, $G_{pq}^{\text{un}} = \langle a_p, a_q \rangle$, so that $c^T G^{\text{un}} c = E_{\text{spec}}$. The off-diagonal interaction matrix is $G^{\text{un}} - D_a$, where $D_a = \text{diag}(\|a_p\|^2)$; and $E_{\text{str}}(\kappa) := \sum_{p \leq \kappa} c_p^2 \|a_p\|^2$, with $\|a_p\|^2 := \sum_k e^{-\varepsilon^2 \gamma_k^2} \sin^2(\gamma_k \log p)$ (Paper-4 fixed phasor convention, no σ -factor; this connects with Paper 4's energy-asymmetry framework and differs from the Paper-5 reference operator $\Phi(\sigma)$ of §2). The reference value is $\eta_{\text{ren}}(53) = 0.66927$; on the tested grid $\kappa \in \{23, 53, 101, 199, 503, 1009\}$ the values lie in $[0.669, 0.820]$ [numerical; no convergence theorem]. The convergence $E_{\text{str}}(\kappa) \rightarrow E_{\text{str}}(\infty)$ follows from the bound $\|a_p\|^2 \leq A(\varepsilon, N)$ and

$$f_p = \frac{4(\log p)^2(2p-1)}{p(p-1)^2} = O\left(\frac{(\log p)^2}{p^2}\right),$$

so $E_{\text{str}}(\kappa) \leq A(\varepsilon, N) \sum_{p \leq \kappa} f_p$ converges absolutely as $\kappa \rightarrow \infty$. The value $E_{\text{str}}(\infty) \approx 6.1$ is numerical at the stated reference parameters [Numerical].

The finite-grid values of $\eta_{\text{ren}}(\kappa)$ admit an exact algebraic description. Assume $\|a_p\| > 0$ for all active primes (satisfied numerically throughout). Set

$$K_\kappa := D_a^{-1/2} G^{\text{un}} D_a^{-1/2}, \quad v_\kappa := D_a^{1/2} c / \sqrt{E_{\text{str}}(\kappa)}.$$

Then $\|v_\kappa\| = 1$ and, writing $K_\kappa = \sum_j \lambda_j u_j u_j^T$,

$$\nu_\kappa^{(w)} := \sum_j |\langle v_\kappa, u_j \rangle|^2 \delta_{\lambda_j}, \quad \eta_{\text{ren}}(\kappa) = 1 - m_1\left(\nu_\kappa^{(w)}\right) \quad \text{exactly.}$$

The unweighted empirical eigenvalue measure $\nu_\kappa := (1/N) \sum_j \delta_{\mu_j}$ gives $m_1(\nu_\kappa) = (1/N) \text{Tr}(\tilde{T}(\kappa))$, which is not identifiable with $c^T G^{\text{un}} c / E_{\text{str}}$; it is included only for comparison.

Observation 4.1 (Numerical). *For $\kappa \leq 1009$, $\varepsilon = 0.05$, $N = 100$:*

$$\eta_{\text{ren}}(\kappa) \rightarrow 0.81 \quad [\text{numerical extrapolation; limit not proved}].$$

The open problem is tightness and convergence of $\nu_\kappa^{(w)}$ and identification of its limiting first moment (Open Problem 6.4). The extrapolated value near 0.81 is not identified with any classical constant ($\eta_\infty \neq \pi^2/6$, $\eta_\infty \neq \gamma_E$).

The rank-one update structure in the cutoff variable deserves comment. In the Paper-5 convention $\sigma = \frac{1}{2}$, each step $\kappa \rightarrow \kappa'$ (adding the next prime p') takes the form

$$\tilde{T}(\kappa') = \tilde{T}(\kappa) + \Phi(\tfrac{1}{2})_{p'} \otimes \Phi(\tfrac{1}{2})_{p'}^T,$$

where $\Phi(\tfrac{1}{2})_{k,p'} := e^{-\varepsilon^2 \gamma_k^2 / 2} \sin(\tfrac{1}{2} \gamma_k \log p')$ is the column of $\Phi(\tfrac{1}{2})$ associated with p' . This rank-1 perturbation structure provides a tractable framework for studying the $\kappa \rightarrow \infty$ limit.

Remark (Relation to the energy-gap structure of Paper 4). *The oscillatory term $O(\sigma)$ at $\sigma = \frac{1}{2}$ shares its trigonometric kernel with the prime-phasor interference structure of Paper 4 [4]. Specifically,*

$$D_{\text{SEL}} - O(\tfrac{1}{2}) = \sum_{p,k} w_k \sin^2(\tfrac{1}{2} \gamma_k \log p)$$

equals the diagonal of $\tilde{T}(\tfrac{1}{2})$, while the energy gap Δ of Paper 4 measures the off-diagonal pairwise prime-phasor interference. Both involve related prime-phasor kernels built from $\gamma_k \log p$, but with different normalisations: the present paper uses the σ -dependent half-angle trace convention at $\sigma = \frac{1}{2}$, while Paper 4 uses the fixed phasor convention without σ .

5 Smoothed Zero Sums and the Bias Conjecture

5.1 Objects and Decomposition

We introduce two analytically distinct objects and establish their relationship to the curvature-bias quantity.

Definition 5.1 (Finite truncation). *For a prime $p \leq \kappa$, set*

$$Z_p := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} p^{i\gamma_k} \in \mathbb{C}, \quad \operatorname{Re}(Z_p) = \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p).$$

Definition 5.2 (Guinand–Weil object and Ordinate Proxy). *For a prime p and $\varepsilon > 0$, define the full Guinand–Weil zero sum*

$$\tilde{Z}_p^{\text{GW}}(\varepsilon) := \sum_{\rho} h_{p,\varepsilon}^{\text{even}}\left(\frac{\rho - \frac{1}{2}}{i}\right),$$

where the sum ranges over all non-trivial zeros ρ of $\zeta(s)$, and define the finite positive-ordinate half-sum

$$Z_{p,N}^+(\varepsilon) := \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p) = \operatorname{Re} Z_p(\varepsilon),$$

restricted to the first N positive ordinates γ_k . The test function and its Fourier transform are

$$h_{p,\varepsilon}^{\text{even}}(u) := e^{-\varepsilon^2 u^2} \cos(u \log p), \quad \hat{h}_{p,\varepsilon}(x) := \frac{\sqrt{\pi}}{2\varepsilon} \left[\exp\left(-\frac{(2\pi x - \log p)^2}{4\varepsilon^2}\right) + \exp\left(-\frac{(2\pi x + \log p)^2}{4\varepsilon^2}\right) \right].$$

In this paper, all computations use $Z_p(\varepsilon)$ (Definition 5.1) and the infinite analogue

$$Z_p^\infty(\varepsilon) := 2 \sum_{k=1}^{\infty} e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p).$$

The full Guinand–Weil object $\tilde{Z}_p^{\text{GW}}$ serves as formal motivation for the sign of Z_p^∞ via its explicit-formula main term (Structural Reduction below), but is not used in computations or the Bias Conjecture. The bare symbol $\tilde{Z}_p$ is not used in this paper.

Equivalently,

$$Z_p^\infty(\varepsilon) = 2 \lim_{N \rightarrow \infty} Z_{p,N}^+(\varepsilon),$$

whenever the Gaussian-damped ordinate sum is taken over all positive ordinates. The relationship between $\tilde{Z}_p^{\text{GW}}$ and Z_p^∞ requires knowledge of off-critical zeros. If all non-trivial zeros lie on the critical line, then by evenness of $h_{p,\varepsilon}^{\text{even}}$,

$$\tilde{Z}_p^{\text{GW}}(\varepsilon) = Z_p^\infty(\varepsilon)$$

in the infinite-ordinate limit. Without this assumption,

$$\tilde{Z}_p^{\text{GW}}(\varepsilon) = Z_p^\infty(\varepsilon) + E_p^{\text{off}}(\varepsilon),$$

where E_p^{off} collects contributions from zeros off the critical line. Controlling this term is Open Problem 6.5.

The factor two here is a zero-side convention: $Z_{p,N}^+$ sums only over positive ordinates, while the full even zero sum counts the paired ordinates $\pm\gamma_k$. Under critical-line localization and in the infinite-ordinate limit, evenness gives $\tilde{Z}_p^{\text{GW}} = 2 \lim_N Z_{p,N}^+ = Z_p^\infty$. The Fourier-transform normalization affects the prime-side constants in the explicit formula; it is recorded separately in the Guinand–Weil convention below.

Remark (Relation to Mazhouda’s asymptotic sums). *The smoothed zero sum $Z_p^\infty(\varepsilon)$ is formally a prime-indexed complex specialisation of Mazhouda’s Selberg-class Gaussian sum $\sum_\rho e^{u\rho^2 - v\rho}$ [8] under $u = \varepsilon^2$, $v = i \log p$. Mazhouda establishes asymptotic behaviour for $v \in \mathbb{R}$ (cases $v = u$ and $v = 0$) [8]; the prime-indexed imaginary- v case and the sign of $\text{Re}(Z_p^\infty)$ are not covered by [8] and constitute the central open problem of this paper.*

The curvature-bias quantity is defined as

$$B := \sum_{k=1}^N \sum_{p \leq \kappa} e^{-\varepsilon^2 \gamma_k^2} (\gamma_k \log p)^2 \cos(\gamma_k \log p).$$

Definition 5.3 (γ^2 -weighted phasor sum).

$$Z_p^{(2)} := \sum_{k=1}^N \gamma_k^2 e^{-\varepsilon^2 \gamma_k^2} p^{i\gamma_k}.$$

The γ_k^2 -weighted phasor sum $Z_p^{(2)}$ carries the frequency weights needed for the curvature decomposition. It is distinct from the unweighted sum Z_p (Definition 5.1), which enters the Bias Conjecture. The γ_k^2 -weighting in the curvature operator is consistent with the UV prolate spectral weights studied by Connes and Moscovici [15] in the context of prolate-function approximations to the zero ordinates; the analytic significance of this weighting remains open.

Proposition 5.4 (Decomposition, proved).

$$B = \sum_{p \leq \kappa} (\log p)^2 \text{Re}(Z_p^{(2)}).$$

Proof. Expand $\text{Re}(Z_p^{(2)}) = \sum_k \gamma_k^2 e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p)$, multiply by $(\log p)^2$, and exchange the order of summation. The product $\gamma_k^2 \cdot (\log p)^2 = (\gamma_k \log p)^2$ recovers the definition of B . $\square$

Remark. *The unweighted decomposition $\sum_{p \leq \kappa} (\log p)^2 \text{Re}(Z_p) = -42.21$ at reference parameters is a distinct quantity, denoted B_{int} , the unweighted integrated bias (ordinate proxy; § 5.3). Both $B = -19342.5$ and $B_{\text{int}} = -42.21$ are negative; the curvature value B is the one entering the $\sigma = \frac{1}{2}$ selection argument via $O''(\frac{1}{2}) = -2B$.*

Numerically, $B = -19342.5$ ($\kappa = 53$, $\varepsilon = 0.05$, $N = 100$, numerical). Numerically, $\operatorname{Re}(Z_p) < 0$ holds for 14 of the 16 primes $p \leq 53$; the two exceptions are $p = 37$ and $p = 53$ ($\varepsilon = 0.05$, $N = 100$).

5.2 Structural Reduction

We apply the Guinand–Weil explicit formula to $h_{p,\varepsilon}^{\text{even}}$ to formally identify the candidate main term of $\tilde{Z}_p^{\text{GW}}(\varepsilon)$. This formal calculation motivates (but does not prove) the sign of $Z_p^\infty(\varepsilon)$.

Convention (Guinand–Weil normalization). We use the explicit formula in the following form. For an admissible even test function h with Fourier transform $\hat{h}(x) := \int_{-\infty}^{\infty} h(u) e^{-2\pi i u x} du$, the Guinand–Weil formula reads

$$\sum_{\rho} h\left(\frac{\rho - \frac{1}{2}}{i}\right) = -\frac{1}{2\pi} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \left[\hat{h}\left(\frac{\log n}{2\pi}\right) + \hat{h}\left(-\frac{\log n}{2\pi}\right) \right] + (\Gamma\text{-terms}) + (\text{constant terms}).$$

The factor $1/(2\pi)$ in front of the prime-power sum reflects the chosen Fourier-transform normalization. This convention is consistent with Weil [7] and Guinand [6].

Structural Reduction 5.2 (formal calculation: formal isolation of candidate main term; dominance open).

Step 1 — Explicit formula template. Applying the convention above to $h = h_{p,\varepsilon}^{\text{even}}$:

$$\tilde{Z}_p^{\text{GW}}(\varepsilon) = -\frac{1}{2\pi} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \left[\hat{h}_{p,\varepsilon}\left(\frac{\log n}{2\pi}\right) + \hat{h}_{p,\varepsilon}\left(-\frac{\log n}{2\pi}\right) \right] + (\Gamma\text{-terms}) + (\text{constant terms}).$$

Step 2 — Fourier transform. For $h_{p,\varepsilon}^{\text{even}}(u) = e^{-\varepsilon^2 u^2} \cos(u \log p)$, the Fourier transform is a sum of two Gaussians:

$$\hat{h}_{p,\varepsilon}(x) = \frac{\sqrt{\pi}}{2\varepsilon} \left[\exp\left(-\frac{(2\pi x - \log p)^2}{4\varepsilon^2}\right) + \exp\left(-\frac{(2\pi x + \log p)^2}{4\varepsilon^2}\right) \right].$$

At the peak $x = (\log p)/(2\pi)$ the two summands contribute $(\sqrt{\pi}/2\varepsilon)(1 + e^{-(\log p)^2/\varepsilon^2})$, where the second term is exponentially suppressed for small ε .

Step 3 — Prime-power localization. For small ε , the Gaussian $\hat{h}_{p,\varepsilon}$ is sharply concentrated near $x = (\log p)/(2\pi)$. In the prime-power sum $\sum_{n \geq 2} (\Lambda(n)/\sqrt{n}) \hat{h}(\log n/2\pi)$, the contribution from $n = p$ (with $\Lambda(p) = \log p$) is the *formally isolated peak contribution*: other prime powers $n \neq p$ contribute $\hat{h}_{p,\varepsilon}(\log n/2\pi)$, which is exponentially suppressed when $|\log n - \log p| \gg \varepsilon$; the quantitative dominance over the local cluster around $\log p$ is not established here and constitutes part of the Bias Conjecture.

Step 4 — Candidate main term. The $n = p$ contribution, using the convention above, is

$$-\frac{1}{2\pi} \frac{\log p}{\sqrt{p}} \left[\hat{h}_{p,\varepsilon} \left(\frac{\log p}{2\pi} \right) + \hat{h}_{p,\varepsilon} \left(-\frac{\log p}{2\pi} \right) \right] = -\frac{\log p}{2\sqrt{\pi} \varepsilon \sqrt{p}} \left(1 + e^{-(\log p)^2/\varepsilon^2} \right).$$

The dominant contribution uses the leading Gaussian peak; the factor $e^{-(\log p)^2/\varepsilon^2}$ is exponentially suppressed for small ε . This candidate main term is strictly negative for all primes $p > 1$. This is a formal statement about $\tilde{Z}_p^{\text{GW}}$; its transfer to Z_p^∞ is conditional on the GW bridge (Open Problem 6.5).

Step 5 — Remainder. All remaining contributions — prime powers $n = p^k$ with $k \geq 2$, other primes, the archimedean Γ -factor functional attached to $h_{p,\varepsilon}^{\text{even}}$, constant terms, and trivial-zero contributions — are grouped into the (Γ -terms) and correction terms. These are real-valued. Their combined magnitude relative to the candidate main term is not controlled quantitatively here.

In summary, the Structural Reduction formally identifies

$$\tilde{Z}_p^{\text{GW}}(\varepsilon) = -\frac{\log p}{2\sqrt{\pi} \varepsilon \sqrt{p}} + (\Gamma\text{-terms}) + (\text{correction terms}).$$

The candidate main term is strictly negative. Whether it dominates quantitatively is the content of the Bias Conjecture (Conjecture 5.7).

Remark. *The Γ -terms and correction terms in the Structural Reduction are real-valued. Their magnitude relative to the candidate main term $-(\log p)/(2\sqrt{\pi} \varepsilon \sqrt{p})$ is not controlled quantitatively in this paper. Establishing this dominance for the full Guinand–Weil object is the explicit-formula part of the Bias Conjecture program; transferring it to Z_p^∞ additionally requires the GW bridge of Open Problem 6.5.*

The following lemma records a formally trivial but logically useful reduction.

Lemma 5.5 (Reduction Lemma, proved). *If $B = -M + E$ with $M > 0$ and $|E| < M$, then $B < 0$.*

Proof. Since $|E| < M$, one has $B = -M + E \leq -M + |E| < 0$. □

This lemma isolates the local dominance step: once a negative main term and a smaller remainder have been identified for the relevant quantity, its sign is fixed. The remaining analytic content lies in proving such dominance, and in the separate bridge and stationarity conditions stated below.

5.3 Sign Transfer and the Bias Conjecture

Definition 5.6 (Finite half-sum and truncation tail).

$$Z_{p,N}^+(\varepsilon) := \operatorname{Re} Z_p(\varepsilon) = \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p),$$

$$R_{p,N}^\infty(\varepsilon) := 2Z_{p,N}^+(\varepsilon) - Z_p^\infty(\varepsilon) = -2 \sum_{k>N} e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p),$$

so that

$$Z_{p,N}^+(\varepsilon) = \frac{1}{2}(Z_p^\infty(\varepsilon) + R_{p,N}^\infty(\varepsilon)).$$

The tail satisfies $|R_{p,N}^\infty(\varepsilon)| \leq 2 \int_{\gamma_N}^\infty e^{-\varepsilon^2 t^2} dN(t) \lesssim 10^{-61}$ at $N = 100$, $\varepsilon = 0.05$, uniformly for $p \leq 53$ [Numerical: Riemann–von Mangoldt density argument, verified by `verify_paper5.py`].

Conjecture 5.7 (Bias Conjecture, open). For each fixed prime p , there exists $\varepsilon_0(p) > 0$ such that

$$Z_p^\infty(\varepsilon) < 0 \quad \text{for all } 0 < \varepsilon < \varepsilon_0(p), \quad Z_p^\infty(\varepsilon) = 2 \sum_{k=1}^\infty e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p).$$

Terminological note. The word “bias” is used here for the sign asymmetry of a smoothed spectral sum at an individual prime scale p . This is not a prime-number race in the sense of Chebyshev or Rubinstein–Sarnak [13], which concerns the comparative distribution of primes in residue classes under GRH. It is also not a Weil positivity statement or a Li-coefficient criterion, both of which are global positivity conditions over all zeros and test functions simultaneously. Conjecture 5.7 is a pointwise spectral-sign claim for a single-prime-indexed observable under Gaussian damping.

The evidence for Conjecture 5.7 is threefold. Structurally, the Structural Reduction of §5.2 formally identifies the candidate main term $-(\log p)/(2\sqrt{\pi} \varepsilon \sqrt{p}) < 0$. Numerically, $\operatorname{Re}(Z_p) < 0$ holds for 14 of 16 primes $p \leq 53$ [Numerical]. As a structural precedent, Mazhouda [8] derived Gaussian-smoothed asymptotic formulas for sums over zeros of L -functions in the Selberg class: for $m \geq 2$,

$$\sum_{\rho} e^{u\rho^2 + (\log m)\rho} = -\frac{\Lambda(m)}{\sqrt{4\pi u}} + O_m(1), \quad u \rightarrow 0^+,$$

showing that Gaussian smoothing isolates prime-power contributions and produces an explicit negative main term. This provides a structural precedent for the negative bias in smoothed zero sums at prime scales. The test functions differ from those used here; a direct application to $Z_p^\infty(\varepsilon)$ requires showing that the present kernel falls within Mazhouda’s framework.

The conjecture remains open because the quantitative dominance of the candidate main term over all Γ -terms and correction terms is not formally established.

We write

$$B_{\text{int}}(\kappa, N, \varepsilon) := \sum_{p \leq \kappa} (\log p)^2 Z_{p,N}^+(\varepsilon) = \sum_{p \leq \kappa} (\log p)^2 \operatorname{Re} Z_p(\varepsilon).$$

Proposition 5.8 (Conditional sign transfer). Assume:

- (A) $Z_p^\infty(\varepsilon) < 0$ for all primes $p \leq \kappa$ (Bias Conjecture);

(B) $|R_{p,N}^\infty(\varepsilon)| < |Z_p^\infty(\varepsilon)|$ for all primes $p \leq \kappa$ (truncation tail strictly dominated by main term).

Then $Z_{p,N}^+(\varepsilon) = \frac{1}{2}(Z_p^\infty + R_{p,N}^\infty) < 0$ for all $p \leq \kappa$, i.e. $\operatorname{Re}(Z_p) < 0$, and consequently

$$B_{\text{int}} = \sum_{p \leq \kappa} (\log p)^2 Z_{p,N}^+ = \frac{1}{2} \sum_{p \leq \kappa} (\log p)^2 (Z_p^\infty + R_{p,N}^\infty) < 0.$$

Proof. By (A), $Z_p^\infty < 0$. By (B), $|R_{p,N}^\infty| < |Z_p^\infty|$. Hence $Z_{p,N}^+ = \frac{1}{2}(Z_p^\infty + R_{p,N}^\infty) < 0$, i.e. $\operatorname{Re}(Z_p) < 0$. Since $(\log p)^2 > 0$, the sum $B_{\text{int}} < 0$. In the infinite positive-ordinate limit, assumption (A) gives directly

$$B_{\text{int}}^{\infty,+} := \frac{1}{2} \sum_{p \leq \kappa} (\log p)^2 Z_p^\infty(\varepsilon) < 0,$$

since κ is finite and each weight $(\log p)^2$ is positive. □

Remark. The curvature quantity $B = \sum_{p \leq \kappa} (\log p)^2 \operatorname{Re}(Z_p^{(2)})$ (Proposition 5.4) carries the additional γ_k^2 weights. Since $\operatorname{Re}(Z_p) < 0$ does not in general imply $\operatorname{Re}(Z_p^{(2)}) < 0$, the Bias Conjecture alone does not give $B < 0$. The stronger statement $B = -19342.5 < 0$ is confirmed numerically at reference parameters and constitutes independent evidence for the Curvature Bias.

Assumption (A) is the Bias Conjecture (Open Problem 6.1). Assumption (B) requires quantitative truncation-error control via the bound of Definition 5.6, which is not established here. Neither condition is verified in this paper.

The logical chain from Conjecture 5.7 to the Curvature Bias at $\sigma = \frac{1}{2}$ proceeds as follows (all conditional). Under the Bias Conjecture and the truncation control of (B), Proposition 5.8 gives $B_{\text{int}} < 0$. The full curvature quantity $B < 0$ is confirmed numerically ($B = -19342.5$ at reference parameters). Note that $B_{\text{int}} < 0$ does not imply $B < 0$: $B = \sum_p (\log p)^2 \operatorname{Re}(Z_p^{(2)})$ carries additional γ_k^2 -weights absent from B_{int} . The missing link is Open Problem 6.3 below. By Reduction Lemma 5.5, $B < 0$ establishes the negative curvature bias. Complementing this with analytic control of the stationarity condition $|O'(\frac{1}{2})| \ll A(\varepsilon, N) \cdot \pi(\kappa)$ (Open Problem 6.2) would then complete the $\sigma = \frac{1}{2}$ Curvature Bias argument, subject to all stated auxiliary conditions.

6 Open Problems

We list five open problems in decreasing order of centrality to the program of this series.

Open Problem 6.1 (Bias Conjecture). *For each fixed prime p , prove that $Z_p^\infty(\varepsilon) < 0$ for all sufficiently small $\varepsilon > 0$, where $Z_p^\infty(\varepsilon) = 2 \sum_{k=1}^{\infty} e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p)$. This is the central open problem of this paper. Its proof via quantitative estimates on the Guinand–Weil explicit formula — controlling Γ -terms and correction terms relative to the candidate main term $-(\log p)/(2\sqrt{\pi}\varepsilon\sqrt{p})$, together with the GW bridge of Open Problem 6.5 if the proof proceeds through the full Guinand–Weil object rather than directly through the ordinate proxy Z_p^∞ , —*

would, in conjunction with the truncation control of Proposition 5.8(B), the Weighted Bias Bridge of Open Problem 6.3 (or an independent analytic proof that $B < 0$), and the Stationarity condition below, complete the $\sigma = \frac{1}{2}$ Curvature Bias argument.

Open Problem 6.2 (Stationarity). Fix $\varepsilon > 0$ and $N \geq 1$, and set $A(\varepsilon, N) = \sum_{k=1}^N e^{-\varepsilon^2 \gamma_k^2}$ as in Theorem 3.1. Establish whether there exists a constant $C_{\varepsilon, N}$, independent of κ , such that

$$|O'(\tfrac{1}{2}; \kappa, \varepsilon, N)| \leq C_{\varepsilon, N} A(\varepsilon, N) \pi(\kappa)$$

for all prime cutoffs κ . At the reference parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $O'(\frac{1}{2}) = +2.475$, approximately 11% of $A(\varepsilon, N)\pi(\kappa)$ [Numerical]. An analytic proof would complement the Bias Conjecture. The quantitative bounds of Carneiro, Chandee, and Milinovich [16] on $S(t)$ provide a natural analytic framework for future control of this bound.

Open Problem 6.3 (Weighted Bias Bridge). Define $Z_{p,2}^\infty(\varepsilon) := 2 \sum_{k=1}^\infty \gamma_k^2 e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p)$. Determine whether the sign implication

$$Z_p^\infty(\varepsilon) < 0 \implies Z_{p,2}^\infty(\varepsilon) < 0$$

holds for sufficiently small $\varepsilon > 0$ at fixed prime p . If the implication fails in general, identify additional hypotheses under which the weighted sign transfer holds, or construct a counterexample mechanism showing that the unweighted and γ_k^2 -weighted signs may decouple. This “weighted bias bridge” is the missing link between the Bias Conjecture (which controls B_{int}) and the curvature quantity B (which carries γ_k^2 -weights and enters $O''(\frac{1}{2}) = -2B$).

Open Problem 6.4 (Weighted spectral measure). Assume $\|a_p\| > 0$ for every active prime $p \leq \kappa$ (numerically satisfied throughout the tested range); equivalently, restrict to the support $S_\kappa^* := \{p \leq \kappa : \|a_p\| > 0\}$. Let $K_\kappa := D_a^{-1/2} G^{\text{un}} D_a^{-1/2}$ and $v_\kappa := D_a^{1/2} c / \sqrt{E_{\text{str}}(\kappa)}$. Then $\|v_\kappa\| = 1$ and $\eta_{\text{ren}}(\kappa) = 1 - \langle v_\kappa, K_\kappa v_\kappa \rangle$ exactly. Let $K_\kappa = \sum_j \lambda_j u_j u_j^T$; define $\nu_\kappa^{(w)} := \sum_j |\langle v_\kappa, u_j \rangle|^2 \delta_{\lambda_j}$. Then $\eta_{\text{ren}}(\kappa) = 1 - m_1(\nu_\kappa^{(w)})$ exactly. The open problem is: establish tightness and convergence of $\nu_\kappa^{(w)}$ as $\kappa \rightarrow \infty$ and identify the limiting first moment.

Open Problem 6.5 (GW bridge). Establish whether the ordinate proxy $Z_p^\infty(\varepsilon) = 2 \sum_{k=1}^\infty e^{-\varepsilon^2 \gamma_k^2} \cos(\gamma_k \log p)$ agrees with the full Guinand–Weil object $\tilde{Z}_p^{\text{GW}}(\varepsilon)$ to leading order without assuming that all non-trivial zeros lie on the critical line.

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. At no point is the Riemann Hypothesis assumed. The ordinates γ_k are used as numerically specified inputs. No property of their distribution requiring RH is assumed or used.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. The operator $\tilde{T}(\sigma)$ is constructed

explicitly from the coupling map $\Phi(\sigma)$ (Definition 2.2). No random matrix hypothesis, no GUE pair-correlation conjecture, and no Hilbert–Pólya postulate enters the construction or the proofs.

$\tilde{T}(\sigma)$ is not postulated. It is derived as $\Phi(\sigma) \circ \Phi(\sigma)^*$ from an explicit matrix. It is not a hypothetical object.

$\tilde{T}$ is not a Hilbert–Pólya operator. The numerical spectral diagnostic in § 2.2 observes $\mu_j \approx C_T/\gamma_{k(j)}$ on the tested grid, hence gives no support for a Hilbert–Pólya interpretation; in any case $\tilde{T}(\sigma)$ is a finite-rank Gram-type operator built from external ordinate inputs, not an operator postulated to have spectrum $\{\gamma_k\}$.

Theorem 3.1 is purely algebraic. The trace formula is proved by a two-line trigonometric manipulation. No asymptotic analysis and no analytic number theory are used.

The Bias Conjecture is open. Conjecture 5.7 is clearly marked as an open problem. No theorem in this paper implies it.

γ_k as inputs, not conclusions. The ordinates appear as numerically specified inputs to the model. The claim $\text{Re}(\rho_k) = \frac{1}{2}$ is not used anywhere; the reference point $\sigma_0 = \frac{1}{2}$ is chosen as the object of study, not assumed to be the unique zero location.

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Call for Feedback

This paper is part of an open research initiative on prime–zero coupling and explicit-formula operators. All papers are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

The author welcomes critical feedback, corrections, and collaboration.

Zenodo:

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GitHub:

<https://github.com/utehrani/analysislab-nt>

Topics of particular interest include: analytical approaches to the Bias Conjecture via Guinand–Weil estimates; connections to existing results on exponential sums over primes and zeros; and any errors or gaps in the present arguments.

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